



Investigation report

Grounding on Boolarong Bank, Merrandi Channel, Kanooka Heads - AAIB-style causal and contributory factors report

Drafted 08/10/2026

INVESTIGATION WORKFLOW SUITE

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Synopsis

On 07/07/2026 MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers, was inbound through the Merrandi Channel at Kanooka Heads on service TF18. It was dark, and 38 passengers, a master, a deckhand and a kiosk attendant were on board. At 18:07:06 the chart plotter display froze and did not recover. Over about a minute the heading eased from 195 to 200 degrees and was then held. The vessel continued at about eight knots and left the dredged channel on its western side at about 18:09. The shallow alarm sounded 16 seconds before contact. At 18:12:04 the vessel grounded on Boolarong Bank at 6.8 kt, 54 m outside the dredged channel, on a falling tide.

There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. The starboard hull bottom plating was set up over 4.1 m, with a 180 mm split in way of the No. 2 void, which flooded to approximately 260 mm. The starboard propeller blades and shaft were bent, the rudder stock was distorted and the skeg was torn away. The master stopped both engines and made no attempt to refloat, which very likely reduced risk by keeping a holed vessel on the bank. All passengers were ashore by 19:26. The vessel refloated at 00:20 and was withdrawn from service, and the channel was closed to commercial traffic for 19 hours.

The investigation found that the plotter freeze and the transit continuing at speed without a position fix very likely caused the grounding. The heading change was steered without any decision to turn, and it took the vessel out of the channel. The shallow alarm's 2019 setting of 0.9 m very likely gave less warning than the vessel needed to stop. The No. 4 beacon, the only remaining reference, was screened by mangroves between 0.84 nm and 0.34 nm north of it. This likely allowed the held heading to go unnoticed, although the turn began at the frozen plotter before any search for the beacon. Clearing of that sightline had very likely been prevented by its deferral, unfunded, at three works rounds. The operator had no night passage plan and no hold point for loss of the primary fix, and this very likely contributed. TF18 had likely been moved into darkness without the daylight-only risk assessment being reopened, although why the change was assessed as a timetable change only was not established. The operator's defect procedure did not require a fault that had cleared itself to be recorded. This likely left the plotter fault uncorrected, although how quickly a recorded freeze would have led to the fix was not established.

Several other matters did not influence this occurrence but increased risk. These were the absence of any notice or availability duty for aids off declared channels, the halving of the beacon's range without notice to channel users, a roster placing the darkness transit at the end of a long split duty, a forward saloon lifejacket locker blocked by a 2019 counter, and the verbal handover of the plotter fault. The investigation could not determine whether shore floodlights behind the beacon delayed its sighting, or whether the master was fatigued. The Merrandi Islet canopy was trimmed on 24/07/2026, local notice LNM-2026-14 was issued and the operator reissued its passage plan. Recommendations were made for a sightline test in the services agreement and for a night passage plan with a hold point.

Factual information

History of the occurrence

At about 13:05 on 07/07/2026, at the Kanooka Heads town wharf, the morning master handed MV Tarrakin Star over to the relieving master. The outgoing master said the chart plotter display had frozen twice that morning and recovered by itself each time. The relieving master replied that it had done the same to him the month before. No defect book entry was made for 07/07/2026 before the occurrence, and the afternoon's two round trips were uneventful.

Service TF18, the last of the day, departed Muntari Island at 17:35 with 38 passengers, a master, a deckhand and a kiosk attendant on board. Sunset was at 17:02 and civil twilight ended at 17:29, so the channel transit was made in full darkness. Every evening return had been made in darkness since the winter timetable began on 01/06/2026, and this was the thirty-first.

The vessel entered the Merrandi Channel from the north at 18:03:58. At 18:04:14 the master sent the deckhand from the bridge to collect fares. At 18:05:30 the vessel was on the charted centreline abeam the No. 2 beacon, making 8.4 kt on 195 degrees. At 18:07:06 the chart plotter display froze. The chart stayed on screen, the vessel symbol stopped moving and touch input was not accepted, and this time the display did not recover.

The heading increased from 195 to 200 degrees over about a minute and was held. The master began looking for the No. 4 starboard-hand beacon at 18:08:31, and the vessel crossed the western edge of the dredged channel at about 18:09. He slid a wheelhouse window open at 18:09:40 and sighted the beacon at 18:10:02. Course over the ground then eased to 197 degrees but stayed west of the channel's 195 degrees until contact.

The echo sounder's shallow alarm sounded at 18:11:48, and both throttles were moved to astern together at 18:11:52. At 18:12:04 the vessel made contact with Boolarong Bank at 6.8 kt over the ground, 74 m west of the centreline and 54 m outside the dredged channel. It came to rest within 11 m, heeled 4 degrees to starboard, on a falling tide.

The master stopped both engines at 18:12:31 and did not attempt to refloat. He addressed the passengers at 18:12:40. At 18:13:04 he broadcast a pan pan on VHF channel 16 with the position and persons on board, and asked for the port workboat, which Port Control called out. All passengers were wearing lifejackets by about 18:22. In the forward saloon the deckhand had to reach the locker behind the kiosk servery counter, where three people were standing. The deckhand sounded the No. 2 starboard void at 18:16, found approximately 75 mm of water, and sounded it every ten minutes thereafter. The workboat came alongside at 18:41 and the last passenger was ashore at 19:26.

There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. The vessel refloated on the rising tide at 00:20 on 08/07/2026, was towed to the town wharf and was withdrawn from service. The Merrandi Channel was closed to commercial traffic for 19 hours.

Recorded information

The sources were the Port Control AIS receiver log AIS-2026-0093-R1, the engine monitoring recorder and steering data, the chart plotter event log, the echo sounder alarm log, and bridge camera audio with the VHF channel 12 and 16 recordings, AUD-2026-0093. Recorded times are in UTC, and local time is UTC plus 10:00. The plotter and echo sounder logs were corrected by 2 seconds to the AIS receiver clock.

The AIS track showed the vessel within 3 m of the centreline until 18:07:06. Rate of turn peaked at 6 degrees per minute at 18:07:40 and returned to zero by 18:08:50, after which the vessel made about 20 m of westing a minute. Speed over the ground fell from 8.4 kt to 7.6 kt between 18:07 and 18:11 with no throttle movement. The vessel's positions relative to the No. 4 beacon were:

- 0.68 nm north when the plotter froze
- 0.47 nm north and 12 m west of the centreline when the master said "Right. Where's four"
- 0.25 nm north and 39 m west when he said "There's four. Bring her back."
- 0.03 nm north and 70 m west when the alarm sounded.

From 18:03 to 18:11:52 the engines ran at 1,780 and 1,784 rpm, within 12 rpm of each other, with no alarm, no temperature or pressure excursion and no throttle movement. Steering hydraulic pressure stayed between 96 and 99 bar, within the normal band of 90 to 105 bar, and rudder angle followed helm demand within 1.5 degrees. Astern thrust was developing at contact but had not reached full.

The plotter log recorded watchdog resets under cause code 0x51 at 10:42 and 11:17 local. At 18:07:06 it recorded the display process not responding, with no reset. Power was removed at 18:16.

The echo sounder alarm had been set at 0.9 m under the keel since commissioning in 2019. It sounded once, 16 seconds before contact, in which time the vessel covered 58 m at 7.1 kt. The echo sounder log records that at the surveyed bank gradient "the alarm could not have sounded appreciably earlier at that setting."

The bridge audio recorded the master saying "Oh, come on." at 18:07:11 and "(unintelligible) frozen again." at 18:07:44, and sustained grinding for approximately 9 seconds from contact.

Equipment information

MV Tarrakin Star is a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers. It completed its annual survey on 11/03/2026 with no outstanding conditions and was last slipped in 03/2025. None of the eleven defect book entries in the preceding 90 days concerned the chart plotter, GPS antenna, echo sounder, steering gear, engines or propulsion controls. The maintenance system held no work order, fault report or correspondence on the plotter in 2026.

The plotter is a 12-inch multifunction display fed by a separate GPS antenna unit, running chart software revision 4.11 on firmware issued 03/2024. The operator did not hold the manufacturer's firmware bulletin issued 11/2024 and had no subscription to the manufacturer's bulletins.

The vessel carries 120 adult and 20 child lifejackets in four lockers. The forward saloon locker holds 40 adult jackets behind the kiosk servery counter. The counter was fitted in a 2019 refit under a minor modification notice that did not trigger a re-survey of the safety equipment arrangement. At build in 2011 the approved arrangement had an open bench with the locker face clear. The same counter is fitted to the operator's two other 24 m vessels.

Meteorology

The master described the weather as clear, with a light westerly and no sea in the entrance. The tide was falling at contact. [NOT ESTABLISHED: recorded tidal height and stream at 18:12]

Personnel information

The master had held a sheltered-waters passenger vessel master's certificate since 2012, with 14 years as master, 11 on this route and approximately 4,100 Merrandi Channel transits. His last route check, on 14/02/2026, was satisfactory. He was on day nine of a twelve-day tour, working a split duty of 05:40 to 08:40 and 13:00 to 19:20, and had slept 6 h 15 the night before. He described himself as tired but not slow, which he said was normal for a split duty. He had made twenty six darkness transits before 07/07/2026.

The outgoing master had 9 years as master, 4 on this route, and approximately 1,400 transits. His last route check was on 09/11/2025, and he had worked 05:40 to 13:05 that day.

The deckhand had been with the operator for 7 months, was in a first winter season on the route, and held a basic marine safety certificate and a coxswain's certificate. [NOT ESTABLISHED: kiosk attendant's experience and training]

Organisational and management information

The operator, Tarrakin Ferries, had no night passage plan for the Merrandi Channel. The master's account was that the existing plan stated that night transits were not scheduled. [NOT ESTABLISHED: content and date of the operator's risk assessment for the service] The passage plan names a speed, a course and three beacons but not who is on the bridge, and procedure OPS-3 is silent on a second person there. Both masters and the deckhand said the timetable change of 01/06/2026 came with a new roster and nothing else.

Both masters described a common practice of entering in the defect book only faults present at handover, and passing on cleared faults by word of mouth. The outgoing master believed the plotter had frozen three or four times in the past year without being written up. The company reporting system's "navigation hazard" category had not been used on this route in three years, and no master interviewed knew it existed. Asked why he had never reported the beacon, the master said: "There's no form for the beacon looks dim."

Tests and research

On 12/07/2026 surveyors removed and tested the steering rams, tiller arms, hydraulic system and autopilot. All were serviceable and the system passed a lock-to-lock timing test. On 21/07/2026 an approved electronics contractor bench tested the plotter for 46 hours with the same antenna and cabling, and the fault did not recur. The contractor reported that cause code 0x51 is addressed by the 11/2024 firmware, which had not been applied to the unit.

The slipping survey found:

- starboard bottom plating set up over 4.1 m between frames 6 and 11, with a 180 mm split in way of the No. 2 void, which held about 260 mm of water at refloating
- all three starboard propeller blades bent aft
- the starboard shaft bent 1.4 mm
- the starboard rudder stock distorted 3 degrees
- the starboard skeg torn away.

The port hull, propulsion and steering were undamaged. The bed at the contact point was 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west, a gradient of about 1 in 70. Furrows ran 11 m from first contact on 197 degrees.

On 18/07/2026 a check from the workboat on the inbound track measured the No. 4 beacon's luminous range at 2.6 nm. The light was not visible from the channel between 0.84 nm and 0.34 nm north of the beacon, because the sightline passes through the mangrove canopy on Merrandi Islet, about 240 m north of the beacon. The canopy measured 5.8 m, against 2.1 m in the 2019

aerial survey, and the lantern stands 4.2 m above chart datum. A vessel west of the inbound track sees the light open earlier.

Other information

The No. 4 beacon (Fl G 2.5s) was converted from mains to a solar lantern in 11/2025, reducing its nominal range from 6 to 3 nautical miles. The chart correction was pending, no local notice was issued and no operator was told. The operator's passage plan still recorded 6 M. Work order WO-118, raised 09/2023, recorded the islet's mangroves screening the beacon and was deferred as unfunded at the 2023, 2024 and 2025 works rounds. WO-176, raised 03/2026, recorded the sightline as fully closed. The Authority's file note placed the islet outside the 50 m scope of schedule 2.4 of the services agreement. The monthly observation check is made from the town wharf, which has no line of sight to the islet, and it recorded the beacon as operating every month from 01/2024 to 07/2026. The aids to navigation register has never been provided to Tarrakin Ferries.

Both masters said the beacon had become harder to see over two years, and both referred to new car park floodlights behind it. The master said he could not tell whether those lights were why he could not see the beacon. The investigation recorded no measurement of the shore lighting.

Analysis

External Influences

Aids code set no notice or availability duty for beacons off declared channels

The Waterways Aids to Navigation Code WAN-2 places different duties on the authority responsible for an aid, depending on where the aid stands. Section 4.2 applies only where an aid is established on a declared shipping channel. In that case the responsible authority "must record its availability, must achieve an availability of not less than 99.0 per cent measured annually, and must notify a change to the aid's character, position or range by notice to mariners before the change takes effect." The Merrandi Channel is not a declared shipping channel. Section 4.3 therefore applies to all six of its beacons, and it requires only that the responsible authority "must be able to demonstrate that the aid is inspected at intervals it has determined and recorded."

The services agreement between Kanooka Shire Council and the Kanooka Heads Port Authority sets those intervals. It does not supply the duties that section 4.3 leaves out. Schedule 2 records that "There is no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid." The code's text and the channel's status are documentary and not in dispute. It is virtually certain that no duty to record availability, and no duty to tell the channel's users of a change, applied to the No. 4 beacon.

Section 4.1 is separate and has no channel threshold. It requires every aid to be maintained "so that it performs the function for which it was established." The Authority reads that duty as lit, on station and showing its charted character, and on that reading considers itself compliant. The code does not define the function. Nothing in it says whether a beacon that cannot be seen from part of the channel it marks is performing it. That general duty applied to the No. 4 beacon whatever the channel's status, and it is not part of the gap considered here.

The case for this gap having a bearing on the occurrence rests on two mechanisms. Had section 4.2 applied, the change of nominal range from 6 to 3 nautical miles in 11/2025 would have been notified to the channel's users before it took effect. The Authority would also have been recording the beacon's availability. These are the only two means in the code by which anything about the No. 4 beacon would have reached the people using the channel. Neither existed for it.

Neither mechanism reaches this occurrence. A notice of the range change would have told the operator about a light that was screened by the mangrove canopy on Merrandi Islet over the stretch in which the vessel left the channel. A light whose range is better known is no more visible through that canopy. The finding on the solar conversion, which is treated as a separate safety issue elsewhere in this analysis, fails on the same ground. An availability figure measures whether a light is lit, and the No. 4 beacon was lit throughout. Its monthly observation log recorded it as operating in every month from 01/2024 to 07/2026.

Nor was the screening left in place because of the code. Section 4.1 applied to the beacon regardless of the channel's status. The Authority's own register had recorded the screening since 09/2023, when work order WO-118 was raised. That work order was deferred at three successive works rounds. The Authority's file note against the later work order, WO-176, explains why. The islet lies 240 m from the beacon, beyond the 50 m vegetation scope in schedule 2.4 of the services agreement, so the work competed with boat ramps and pontoons in the general waterways vote. The deferral is argued as a causal factor in its own subsection. It owes nothing to the threshold in section 4.2.

The evidence therefore does not leave the question open. Each route by which the code's threshold might have bore on the grounding runs into a fact that settles against it. It is unlikely that the

absence of notice and availability duties influenced the occurrence, and it is not placed among the factors leading to it.

The gap matters for a different reason. The Merrandi Channel carries a 96-passenger ferry in darkness. Under the code, its operator has no right to be told when a beacon's range, character or position changes, and has no figure for how often the channel's aids work. The solar conversion of the No. 4 beacon showed the effect. No local notice to mariners was issued, no operator using the channel was told, and the register was never provided to Tarrakin Ferries. On the night, the operator's passage plan appendix still recorded the beacon at 6 M. The next change, to a light that is not screened, would reach the channel's users no more readily.

It is likely that this gap increased risk to vessels using the channel. In the terms of this report it is therefore a contributory factor in the sense of a condition that increased risk. It is not one that influenced this grounding. It is a standing gap in what the users of an undeclared channel are told. The Authority can close it by its own practice, whatever the code requires.

The absence of any duty under WAN-2 to notify users of the Merrandi Channel of changes to its aids, or to record their availability, did not influence the grounding. It likely left a standing risk that warrants attention as a safety issue.

Organisational Influences

Last service moved into darkness without reopening the daylight-only risk assessment

The route risk assessment for the Merrandi Channel, RRA-07 issue 2 of 03/2023, carried a single operating restriction: "daylight transits only". It stated that "A night transit of the Merrandi Channel requires the General Manager Operations' approval and a specific passage plan." Change record CHG-2026-014 moved service TF18 from 16:15 to 17:35 for the winter timetable from 01/06/2026. It assessed the change as a "timetable change only", recorded "No new hazard identified" and answered "No" to whether a route risk assessment review was required. The Operations Manager approved it on 22/04/2026, one level below the person the restriction named. The change record does not refer to the restriction. RRA-07 was not reopened, no night passage plan was written, and the General Manager Operations' approval was neither sought nor given. Both documents are current controlled copies, so it is virtually certain that the change was made in this way.

The change asked whether anything about the route was new. It did not ask whether the existing assessment still held for a night operation. A review of RRA-07 against such an operation would have reached the unlit reach between the No. 2 and No. 4 beacons, which the assessment already named as a hazard. Because the restriction itself named the plan, a review that read it could not have closed without a night passage plan. A plan written for this channel at night was the document most likely to say what to do if the primary fix was lost.

The evidence against this link was considered. A reopened assessment might have produced a night plan that said no more than the daylight one, with no hold point, and the vessel would then have met the frozen plotter in the same way. That argument goes to how good the plan would have been, not to whether one would have existed. The investigation did not establish why the change was assessed as a timetable change only. The Operations Manager's understanding of the restriction was given only as a verbal account. That gap affects the explanation of the decision but not its consequence.

It is likely that this decision led to the absence of a night passage plan or hold point for loss of the primary fix, which is argued under Risk Controls, and the evidence puts it as a causal factor.

Clearing of the beacon's sightline deferred unfunded for three works rounds

The Port Authority's register shows that work order WO-118, raised in 09/2023, identified mangrove regrowth on Merrandi Islet screening the No. 4 beacon from the northern approach. It recommended trimming to 2 m on a three-yearly cycle, at an estimated \$18,500. The work was deferred as unfunded at the 2023, 2024 and 2025 works rounds. By 03/2026, WO-176 recorded the sightline as fully closed on the northern approach. The Authority's file note placed the islet, 240 m from the beacon, outside the 50 m scope of schedule 2.4. The work therefore competed with boat ramps and pontoons in the general waterways vote, not with aids to navigation. The deferral is documented and the canopy was measured on site, so its existence is virtually certain.

The step from deferral to screening rests on the Authority's own figures. The canopy stood at 2.1 m in the 2019 aerial survey and at 5.8 m when measured on 18/07/2026, a growth of about half a metre a year. Trimmed to 2 m at the 2023 round, it would have stood near 3.5 m on the night. That is about 0.7 m below the 4.2 m lantern, rather than 1.6 m above it, and the beacon would have been open to the channel over the stretch where it was screened. The only reservation is that the growth may not have been even over seven years, since only two measurements and the work order's description exist between those dates.

It is very likely that the deferral led to the screening of the No. 4 beacon between 0.84 nm and 0.34 nm, which is argued under Local Conditions, and the evidence puts it as a causal factor.

Beacon's range halved by solar conversion with no notice to channel users

In 11/2025 the No. 4 beacon was converted to a solar lantern rated at a nominal 3 nautical miles, against the 6 of the lantern it replaced. A notification reached the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice was issued and no operator was told. The operator's current passage plan appendix still gave the beacon at 6 M.

The case for an effect on this occurrence was examined and did not hold. Over the half mile in which the vessel left the channel the light was screened by the canopy, and a brighter light behind a canopy is still not visible. The master already knew the light was weaker than the mains light it had replaced, so a notice would have told him nothing new. Range was not the limit either: the search ran from 0.47 nm to 0.25 nm, well inside the measured 2.6 nm. Nothing shows that the plan's 6 M figure was used on the night. The one remaining mechanism is the light's conspicuity against the boat ramp floodlights once it cleared the islet. That question is unresolved and is discussed with the floodlights. It is unlikely that the conversion influenced this occurrence.

It remains a safety issue. A light on a 96-passenger route had its range halved without any user being told, and on a night when it is not screened, a master working to the plan would be looking for a light twice as strong as the one there. It is likely that this matters for future safety, which LNM-2026-14 and the reissued appendix now partly address.

Darkness transit rostered to close a split duty within 20 minutes of the spread limit

In the winter timetable, TF18 closes a split duty of 05:40 to 08:40 and 13:00 to 19:20. That is 9 hours 20 minutes of duty across a 13 hour 40 minute spread, against the operator's limits of 10 and 14 hours. The darkness transit therefore falls in the last hour and a quarter of the spread on every day of a twelve-day tour. No relief master is rostered.

The roster could reach this occurrence only through fatigue, and whether the master was fatigued is an unresolved matter discussed under Local Conditions. His recorded performance showed no sign of it. He held the centreline to within 3 m, responded to the shallow alarm in 4 seconds, made complete and correct radio calls at the first attempt, and decided correctly not to refloat. Both

masters gave the same account that no document required a stop when the plotter froze. It is unlikely that the roster influenced this occurrence.

It is nonetheless likely to matter. The roster places every winter darkness transit at the same point in the duty, the fatigue diary the operator asks masters to keep goes unread, and there is no relief master to break a tour. This is a safety issue.

Risk Controls

No night passage plan or hold point for loss of the primary fix

The transit was made under OPS-3 and its appendix A, issue 2 of 02/2023, which is the current issue. Appendix A names the chart plotter as the primary means of position fixing. The secondary means is the "visual transit of the No. 4 beacon and the town wharf leading lights". The appendix refers to darkness only once, in its closing note: "Night transits are not scheduled on this route." It names no hold point or abort criterion. Nothing in it, or anywhere else in the manual, says what a master should do if the plotter is lost in the channel. OPS-3 3.6 calls for reduced speed and consideration of anchoring, but only in reduced visibility, and visibility on the night was not reduced. The operator confirmed that no night passage plan exists, and it is virtually certain that this gap existed.

In practice the secondary means was weaker than the plan suggests. The leading lights lie about 1.9 nm beyond the No. 6 beacon and do not resolve as two lights until a vessel is inside the channel's southern third. The plotter froze 0.68 nm north of the No. 4 beacon. At that point the secondary means therefore rested on the beacon alone. Why the beacon could not be seen at that range is argued under Local Conditions.

The display froze at 18:07:06. The vessel crossed the western edge of the channel at about 18:09, which gave the master 114 seconds in which stopping would have kept the vessel in dredged water. Even after that, a crash stop begun before about 18:11:40 would still have stopped the vessel short of the contact point. The water was flat and there was no other traffic, and a named hold point would have applied throughout. Both masters interviewed said independently that stopping was the right response, and both said that nothing in the operator's documents asks for it.

There is one reservation. A master who did not think to stop unprompted might not have applied a written rule under the same pressure. The investigation weighed this and did not find that it displaced the conclusion.

The gap follows from the decision, argued above, to move TF18 into darkness without reopening RRA-07. RRA-07 itself required "a specific passage plan" for any night transit. The absence of a night passage plan with a hold point for loss of the primary fix very likely allowed the transit to continue at 8 knots from the plotter freeze until the shallow alarm, and it was a causal factor.

Defect procedure never requires a fault that cleared itself to be recorded

OPS-9 decided whether the plotter fault reached anyone ashore. Under 9.1, only a defect that renders the vessel unseaworthy or affects a statutory item must be reported immediately. The plotter is not a statutory item on this vessel on this route, although the passage plan names it as the primary means of position fixing. Under 9.2, all other defects are entered "at the end of the shift in which they are found." That provides no trigger for a fault that has cleared before then, and the manual does not say whether such a fault is a defect at all. Section 9.3 requires a briefing at a change of master but does not require it to be written down. It also requires nothing of the relieving master afterwards.

Both masters, without prompting, described the same reading of the manual. A fault present at handover is written in the book, and one that has cleared is mentioned by word of mouth. The eleven entries in the 90 days before the occurrence bear this out, because each was a fault present when it was written. It is virtually certain that the procedure contained this gap.

On the outgoing master's account, the plotter froze three or four times on this vessel in the year before the occurrence. It froze on the relieving master about a month earlier, and twice on the morning of 07/07/2026. None of these freezes was written down. Had they been recorded, they would have reached the Operations Manager's weekly review and the maintenance system. A single bench test was enough for the contractor to link cause code 0x51 to the 11/2024 firmware, which had not been applied.

Against this, a recorded entry would still have needed someone ashore to find the manufacturer's bulletin, and the operator held no subscription to the bulletins. A recorded freeze might have been reviewed, noted and left. Nothing establishes how many recorded freezes it would have taken for the unit to be sent away, or how quickly. A year of recorded freezes on the primary means of position fixing is unlikely to have gone to nobody, but how quickly any fix would have followed could not be determined.

The verbal handover at 13:05 was one instance of this procedure. It is treated separately as a matter that did not influence the occurrence. The procedure explains that omission but does not excuse it. The absence of any requirement to record a fault that had cleared itself likely left the cause of the plotter freeze uncorrected, and it was a causal factor.

Forward saloon lifejacket locker blocked by a 2019 counter fitted without re-survey

While issuing lifejackets, the deckhand had to ask three passengers to move away from the kiosk counter. The deckhand then passed behind the counter through a 0.6 m gap to reach the forward saloon locker, which holds 40 adult lifejackets.

The survey of 16/07/2026 found that the locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter. With people standing at the counter, the gap is the only route. The muster placard directs passengers to the locker, which cannot be seen from the seats. The arrangement approved at build in 2011 had an open bench in that position. The counter was fitted in a 2019 interior refit under a minor modification notice, which did not trigger a re-survey of the safety equipment arrangement. It is virtually certain that the obstruction existed.

The obstruction played no part in the grounding. All 38 passengers were in lifejackets within about eight minutes and ashore within 74 minutes. The vessel was upright and stable, and the muster was unhurried. Even so, the muster in the forward saloon depended on crew help to reach the jackets. A fire, a night evacuation into the water, or an injured deckhand would have turned that delay into a saloon left waiting for its lifejackets.

The locker remains behind the counter on this vessel. On the operator's account, the same counter is fitted to its two other 24 m vessels. The blocked locker very likely increased risk to passengers, and it is a safety issue that did not influence this occurrence.

Local Conditions

Chart plotter display froze during the channel transit and did not recover

Three independent sources place the freeze at 18:07:06. The plotter's own event log records the display process not responding at that second, with no reset following it. The bridge audio matches it: "Oh, come on" at 18:07:11, repeated screen taps at 18:07:26 and "(unintelligible) frozen again" at 18:07:44. The master's account is that the chart picture stayed on the screen with the vessel

symbol stationary and the display not accepting touch input. The 46 hour bench test on 21/07/2026 did not reproduce the fault and found no defect. That result bears on why the display froze, not on whether it did. The cause code, 0x51, was the same one logged at 10:42 and 11:17 that morning. It is known to the manufacturer and is addressed by firmware issued in 11/2024, which had not been applied to this unit. It is virtually certain that the display froze.

The vessel had held the centreline while the plotter worked. The heading began to change in the second of the freeze, while the master was at the screen. His search for the beacon began 85 seconds later, and no position fix of any kind was taken in between. It is extremely likely that the freeze contributed to the grounding. It was a causal factor, because without it the transit would have continued on the primary means of fixing that the passage plan prescribed.

No. 4 beacon screened by mangroves from the channel, 0.84 to 0.34 nm

The range and visibility check on 18/07/2026 was made on the inbound track at the same time of night. It found that the beacon cannot be seen from the channel between 0.84 nm and 0.34 nm north of it. Over that stretch the sightline passes through the canopy on Merrandi Islet, which measured 5.8 m against a lantern height of 4.2 m. The Port Authority's register had recorded the same screening since 09/2023. It is virtually certain that the screening existed.

With the plotter frozen, the beacon was the only means of fixing left to the master. He began looking for it at 18:08:31, at 0.47 nm, and called it at 18:10:02, at 0.25 nm, 91 seconds later. The vessel crossed the channel's western edge at about 0.40 nm, inside the screened arc. Had the light been visible on the starboard bow, it would have shown the vessel drawing west within a minute of the turn, while the vessel was still in the channel.

Two points weigh against this. The heading change began before the search did, so the screening cannot have caused the turn. It can only have allowed the turn to go unnoticed. In addition, the light was clear of the islet for at least the last 37 seconds of the search, from about 18:09:25, and for longer from the vessel's own track west of the inbound one. For that part of the search the light was there to be seen and was not seen. That later interval is the question taken up under the next heading. It does not undo the screening over the minute and a half before it. It is likely that the screening contributed to the grounding, and it was a causal factor because it removed the only remaining reference while the new heading was held.

Unshielded boat ramp floodlights aligned behind the No. 4 beacon from the north

The foreshore lighting survey found 14 unshielded floodlights on 12 m columns, commissioned on 22/05/2026. Each presented about 4,800 candela toward the channel. From 0.5 nm north of the beacon the array lies within 4 degrees of the same bearing and within 1.5 degrees of the same elevation. It is continuously lit, while the beacon shows a 0.4 second green flash every 2.5 seconds. The existence and alignment of the floodlights are virtually certain.

What the floodlights did on the night could not be settled. In support, the master described looking "into a wall of light". The outgoing master independently described the beacon as "washed out altogether from up the top". The unseen 37 seconds also fall in the interval when the canopy no longer stood in the way. Against this, the beacon was screened by the canopy for most of the search and would not have been visible against a dark background either. The turn began at the plotter, before the search. No measurement exists of what could be seen on the night, and no detection threshold has been established for a short green flash against that background. The master himself said twice that he could not tell whether the lights were the reason he could not see the beacon. Resolving the matter would need a measurement of whether that flash is detectable against the array. The question therefore remains unresolved. The investigation judged it about as likely as not either way, and nothing is carried from it into the conclusions.

Tide set down below prediction over Boolarong Bank

A set-down was very unlikely: the Kanooka Heads gauge read 0.91 m at 18:12 against a predicted 0.94 m, and the vessel would have touched at the predicted height as well, so the hypothesis was excluded.

Master fatigued during the channel transit

Whether the master was fatigued could not be determined, and the investigation judged it about as likely as not. Several facts point towards it. The voyage fell on day nine of a twelve day tour, in the last hour and a quarter of a 13 hour 40 minute spread. The master recorded 6 hours 15 minutes of sleep the night before. In interview he said "I was tired. I'd be lying if I said otherwise." After sighting the beacon he eased back to port rather than stopping, believing himself just outside the edge when he was nearly 40 m west of the centreline.

In the same answer, however, he said "I wasn't fighting to stay awake and I wasn't slow", and the recorded performance supports him. He held the centreline to within 3 m and responded to the shallow alarm within 4 seconds. His radio work was correct at the first attempt, and he decided not to refloat in about thirty seconds. Even had fatigue been established, the decisions that mattered are either accounted for by the absence of a night passage plan and hold point or contradicted by the recorded data. The matter remains unresolved and contributes nothing to the conclusions. Whether the roster placed him in that position is addressed under Organisational Influences.

Individual Actions

Transit continued at eight knots from the plotter freeze until the shallow alarm

After the display froze at 18:07:06, the vessel kept its speed. The AIS track shows 8.4 kt at the freeze and 8.1 kt at 18:08:00. The engine recorder shows no throttle movement until 18:11:52. The later fall to 7.1 kt is explained by the drag of shoaling water.

The bridge audio contains nothing that suggests stopping was considered. At 18:08:31, 85 seconds after the freeze, the master turned to looking for the No. 4 beacon with "Right. Where's four." At 18:10:02 he sighted it with "There's four. Bring her back." He then knew he was west of the line. He believed he was only just outside the channel edge, but he was 39 m west of the centreline. He brought the vessel back to port rather than stopping. The course over ground eased from 200 to 197 degrees, which was still west of the channel's 195.

His account, given in interview and in his safety report, is consistent with this. He did not consider stopping when the plotter froze, because he regarded the visual transit as the backup. After the sighting, stopping on the edge of a bank he could not see felt worse to him than easing back in.

Stopping was available to him throughout this period. There was no other traffic, the water was flat and the wind was offshore at 8 kt. The vessel's crash stop distance from 8 kt is 92 m. The vessel crossed the western edge 114 seconds after the freeze. A stop begun at any point before about 18:11:40 would have halted it short of the contact point. A stop in the first 114 seconds would have left it in the channel. A stop after the sighting would have left it on the margin of the bank, upright and not holed. That window included the 98 seconds after the master knew he was off the line.

The master was alone on the bridge. At 18:04:14 he sent the deckhand to collect fares with "Nah, go and do the fares." This was the normal practice on the route, and nothing in OPS-3 addresses it.

This decision sits downstream of other factors. The passage plan contained no hold point and nothing about losing the plotter, which is argued under Risk Controls. The beacon was screened from the channel over most of the search, which is argued under Local Conditions.

It is extremely likely that continuing at eight knots without a means of fixing position contributed to the grounding, and the evidence places it as a causal factor.

Heading eased five degrees to starboard without a decision to turn

The heading began to change at 18:07:06, the same second the plotter froze. It increased steadily from 195 to 200 degrees over about a minute and was then held. The rate of turn peaked at 6 degrees per minute at 18:07:40 and was back to zero by 18:08:50. During that period the master was occupied with the frozen screen. The audio records screen taps at 18:07:26 and "frozen again" at 18:07:44.

Rudder angle followed helm demand within 1.5 degrees throughout, so the turn was steered. The excluded steering hypothesis is dealt with in Occurrence Events.

The master's account supports the recorded data. In interview he said "I eased her to starboard while I was fiddling with the plotter and then looking for four. Five degrees. I never made that decision." Most of the turn was made at the plotter, before the spoken search began. It was therefore a turn that went unnoticed, rather than one made while looking for the light. Nothing on the bridge showed him the new heading. The vessel symbol on the plotter was frozen, and the beacon was screened.

There was no current and no wind effect at that state of tide, so this heading change alone accounts for the westing. Held at 200 degrees, it moved the vessel west at about 20 m a minute. The vessel crossed the 20 m to the western edge by about 18:09:00.

It is virtually certain that the unnoticed heading change took the vessel out of the dredged channel, and the evidence places it as a causal factor.

Engines stopped and no attempt made to refloat on a falling tide

The master's actions after contact were prompt. He stopped both engines at 18:12:31, 27 seconds after contact, and did not restart them. He addressed the passengers within 36 seconds of the impact. At 18:13:04 his pan pan stated "I am not attempting to refloat" as a decision already made.

He then had the No. 2 starboard void sounded and reported 75 mm of ingress at 18:17:05. The void was sounded every ten minutes after that, and all 38 passengers were issued lifejackets. In interview he gave his reasons as the falling tide, the unknown bottom, the number of people aboard, and the fact that a vessel on the bank cannot sink.

The survey bore out his caution. It found a 180 mm split in the starboard plating in way of that void. Ingress had reached approximately 260 mm by the time the vessel refloated at 00:20. Backing off into 3.2 m of water with the damage unassessed would have put a flooding compartment into deep water with the passengers still aboard.

It is very likely that stopping the engines and not refloating reduced the risk to those on board, and this was a positive action taken quickly.

Plotter fault passed on verbally at the change of master and not entered in the defect book

The plotter's log records freezes at 10:42 and 11:17 that morning. The unit recovered from both by itself. At the handover at about 13:05, the outgoing master told the relieving master about both freezes. The relieving master said the unit had done the same to him the month before. Neither master entered the fault in the defect book or reported it ashore. Their accounts agree, and both volunteered the omission before being asked about it.

It is virtually certain that this omission occurred. It is unlikely, however, that it influenced the occurrence. The freeze at 18:07 was a firmware fault. An entry at 13:05 would have been reviewed at the end of the week under OPS-9. Nothing written that afternoon could have updated the unit or taken the vessel off TF18 by 18:07.

The omission is still likely to be important, because it reflects the practice and not a lapse by two individuals. A cleared fault on the primary means of position fixing that is passed on by word of mouth leaves the next master informed and the shore organisation unaware. It is the instance of the defect procedure gap that is argued under Risk Controls, and it is properly addressed there.

Occurrence Events

Vessel left the dredged channel on the bank side

The question for this part of the analysis is where the vessel was when it grounded, and how it came to be there. The AIS track answers the first part with little room for doubt. At 18:05:30, abeam the No. 2 beacon, the vessel was 1 m west of the charted centreline, and it held the centreline until 18:07:06. From that moment the heading increased steadily from 195 to 200 degrees over about a minute and was then held. The vessel set west at about 20 m per minute. It reached the channel's western edge at about 18:09, 0.40 nm north of the No. 4 beacon. By 18:10:02 it was 39 m west of the centreline, and it was over Boolarong Bank from about 18:11.

The shape of that departure matters as much as its extent. The recorded rate of turn peaked at 6 degrees per minute at 18:07:40 and was back to zero by 18:08:50. That is the signature of a heading change applied and then held. It is not the signature of a vessel being carried off by current or wind. The wind was offshore at 8 kt, and there is no cross-set in the channel at that state of tide. After 18:10:02 the course over the ground eased back to 197 degrees, a turn of about three degrees to port. This slowed the westing but did not stop it, and the course stayed west of the channel's 195 degrees until contact.

The physical evidence on the bed agrees with the track. The grounding position was surveyed at 74 m west of the charted centreline, which is 54 m outside the channel's western edge. Furrows consistent with the starboard hull and skeg ran 11 m from the point of first contact on a heading of 197 degrees.

The channel is 40 m wide and dredged to 3.2 m. The bank on its western side dries in places, and once outside the western edge the vessel was over water shoaling at about 1 in 70. The grounding followed directly from the vessel being outside the dredged channel, and from nothing else. The reasons the heading changed and the speed was maintained belong to the analysis of Individual Actions, where the five degree heading change and the continued transit at eight knots are each argued. Here it is enough to say that both led to this departure.

It was virtually certain that the vessel left the dredged channel on the bank side, and virtually certain that this contributed to the grounding. It is therefore a causal factor.

Shallow alarm at its 2019 setting of 0.9 m gave sixteen seconds' warning

The echo sounder's own log states its setting plainly: "Shallow alarm set to 0.9 m under the keel." That value was written into the setup when the unit was commissioned in 2019 and was never changed. The alarm sounded once on the voyage, at 18:11:48. The bridge audio records the tone beginning in the same second, and the alarm log, the audio and the recorded speed agree to the second. Both throttles were moved to astern together 4 seconds later, at 18:11:52. Contact followed at 18:12:04, 16 seconds after the alarm. Astern thrust was developing at that moment but had not reached full.

At 7.1 kt the vessel covered 58 m in those 16 seconds. The vessel's own trials figure for a crash stop from 8 kt is 92 m. The log is explicit that, on the surveyed gradient of about 1 in 70, "the alarm could not have sounded appreciably earlier at that setting." The equipment therefore did exactly what it had been set to do. The difficulty lay in what that setting could deliver on that slope at that speed, which was a warning shorter than the distance needed to stop.

The surveyed slope also shows what a different setting would have achieved. At about 1.6 m under the keel the alarm would have sounded about 107 m short of the contact. That would have been enough for a crash stop, with four seconds in which to react, and it still would not have sounded in the dredged channel at low water. A setting long enough to give forty seconds' warning would sound in the channel, however. The setting therefore has to be chosen for the channel and the speed at which it is transited, rather than simply raised.

It was virtually certain that the alarm was set at 0.9 m and gave sixteen seconds' warning, and extremely likely that this contributed to the grounding. It is therefore a causal factor. The equipment itself was not at fault.

Steering or propulsion failure before the vessel left the centreline

A failure of the steering or propulsion before the vessel left the centreline was examined and found very unlikely, because the engine recorder showed both engines running within 12 rpm of each other with no alarm, excursion or throttle movement until 18:11:52, steering hydraulic pressure held at 96 to 99 bar within its normal band, the rudder followed helm demand within 1.5 degrees throughout, and every steering component removed and tested on 12/07/2026 was serviceable, with the damage to the starboard shaft, propeller and rudder all dating from the contact.

Conclusions

The investigation identified the following causal and contributory factors. They should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. Under WAN-2, the duties to record an aid's availability, to achieve 99.0 per cent availability and to notify a change of character, position or range by notice to mariners applied only to aids on a declared shipping channel. The Merrandi Channel was not one, so nobody using it had any right to be told of a change to its beacons. (Safety issue)
2. CHG-2026-014 moved service TF18 from 16:15 to 17:35 from 01/06/2026 and assessed this as a timetable change only. It was approved by the Operations Manager on 22/04/2026 without RRA-07 being reopened. RRA-07 restricted the channel to daylight transits and required both the General Manager Operations' approval and a specific passage plan for any night transit. (Safety issue)
3. Why the change was assessed as a timetable change only was not established.
4. Work order WO-118, raised in 09/2023 to trim the mangroves screening the No. 4 beacon, was deferred as unfunded at the 2023, 2024 and 2025 works rounds. The islet lay 240 m from the beacon, outside the 50 m scope of schedule 2.4 of the services agreement, so the work competed for the general waterways vote. (Safety issue)
5. If the trim had been funded at the 2023 round, the canopy would have stood at about 3.5 m on the night, below the 4.2 m lantern, rather than at 5.8 m.
6. The solar conversion of the No. 4 beacon in 11/2025 reduced its nominal range from 6 to 3 nautical miles. No local notice to mariners was issued, the chart correction was still pending on 07/07/2026, and the operator's passage plan still recorded the beacon at 6 M. (Safety issue)
7. The winter roster placed the TF18 darkness transit in the last hour and a quarter of a split duty whose spread of 13 hours 40 minutes ended 20 minutes inside the operator's limit. This applied on every day of a twelve-day tour, and no relief master was rostered. (Safety issue)
8. The operator had no night passage plan for the Merrandi Channel. Its daylight plan named the chart plotter as the primary means of position fixing but gave no hold point, no abort criterion and no guidance on losing the plotter. (Safety issue)
9. OPS-9 did not require a fault that had cleared itself to be recorded. As a result, none of the plotter freezes in the year before the occurrence was ever written down, and the firmware issued in 11/2024 to address cause code 0x51 was never applied to the unit. (Safety issue)
10. The kiosk counter fitted in the 2019 refit under a minor modification notice, without a re-survey, meant the forward saloon lifejacket locker could not be reached from 22 of the 34 forward saloon seats without passing behind the counter. (Safety issue)
11. The chart plotter display froze at 18:07:06 under cause code 0x51 and did not recover. A 46 hour bench test did not reproduce the fault and found no physical defect.
12. The No. 4 beacon was not visible from the channel between 0.84 nm and 0.34 nm north of it, because the sightline passed through the mangrove canopy on Merrandi Islet.

13. Fourteen unshielded LED floodlights at the new boat ramp, commissioned on 22/05/2026, lay behind the No. 4 beacon within 4 degrees of the same bearing from 0.5 nm north. The Port Authority's aids to navigation officer was not consulted on the development.
14. The transit continued at about eight knots from the plotter freeze until the shallow alarm, with no lookout on the bridge. A crash stop begun at any time before about 18:11:40 would have stopped the vessel short of the contact point.
15. From 18:07:06 the heading was eased from 195 to 200 degrees over about a minute and then held. The turn was steered, but the master never decided to make it.
16. The master stopped both engines 27 seconds after contact and made no attempt to refloat on the falling tide, with the vessel holed in way of the No. 2 starboard void.
17. At the change of master at about 13:05, the two morning plotter freezes were passed on by word of mouth and were not entered in the defect book. Both masters volunteered the omission.
18. The vessel left the dredged channel on its western side at about 18:09:00, 0.40 nm north of the No. 4 beacon, and was over Boolarong Bank from about 18:11:00.
19. The shallow alarm, set at 0.9 m under the keel since commissioning in 2019, sounded 16 seconds before contact. In that time the vessel covered 58 m, against a crash stop distance of 92 m from 8 kt.
20. There was no steering or propulsion failure before the vessel left the centreline.

Causal factors

- It is virtually certain that service TF18 was moved into darkness without the daylight-only risk assessment RRA-07 being reopened. It is likely that this contributed to the grounding, because a review of RRA-07 could not have closed without a night passage plan, although why the change was assessed as a timetable change only was not established. (Safety issue)
- It is virtually certain that no night passage plan, and no hold point for loss of the primary fix, existed for the Merrandi Channel. It is very likely that this absence contributed, because it gave the master no prompt to stop when the plotter froze. (Safety issue)
- That the defect procedure never required a fault that cleared itself to be recorded is virtually certain. It likely contributed to the uncorrected plotter fault, although how quickly a recorded freeze would have led to the 11/2024 firmware being applied was not established. (Safety issue)
- The chart plotter display froze during the channel transit and did not recover, which is virtually certain. This very likely contributed by removing the primary means of position fixing at the second the heading began to change.
- It is virtually certain that the transit continued at eight knots from the plotter freeze until the shallow alarm. This very likely contributed, since stopping was available until about 18:11:40, including the 98 seconds after the master knew he was off the line.
- The deferral of the beacon's sightline clearing, unfunded for three works rounds, is virtually certain. It very likely contributed by leaving the canopy above the lantern on the night. (Safety issue)
- The screening of the No. 4 beacon by mangroves from the channel between 0.84 and 0.34 nm is virtually certain. It likely contributed by allowing the held heading to go unnoticed over the minute and a half that mattered, although the turn itself began at the frozen plotter, before the search for the beacon.

Fictional case study for training. Not a real investigation.

- With the heading eased five degrees to starboard without a decision to turn, both the existence of the action and its contribution were virtually certain. With no current or wind effect at that state of tide, it was the sole source of the westing.
- It is virtually certain that the vessel left the dredged channel on the bank side, and virtually certain that this contributed, because the grounding followed directly from the vessel being outside the channel.
- The shallow alarm's 2019 setting of 0.9 m, which gave sixteen seconds' warning, is virtually certain. It very likely contributed, because that warning was shorter than the vessel's stopping distance at seven knots.

Contributory factors

- It is virtually certain that the aids code set no notice or availability duty for beacons off declared channels. It is likely that this increased risk to users of the Merrandi Channel, who had no right to be told of a change to its aids, although it did not influence this occurrence. (Safety issue)
- The halving of the beacon's range by solar conversion, with no notice to channel users, is virtually certain. It likely increased risk, because the operator's passage plan still described a 6 M light. It did not influence this occurrence, because the light was screened over the stretch where the vessel left the channel. (Safety issue)
- It is virtually certain that the darkness transit was rostered to close a split duty within 20 minutes of the spread limit, and likely that this increased risk on every winter darkness transit of the timetable. It did not influence this occurrence, because the master's recorded performance showed no effect of fatigue on the night. (Safety issue)
- The blocking of the forward saloon lifejacket locker by a 2019 counter fitted without re-survey is virtually certain. It very likely increased risk to passengers in any emergency where crew help is not available, although all 38 passengers were in lifejackets within about eight minutes on this occasion. (Safety issue)
- That the plotter fault was passed on verbally at the change of master and not entered in the defect book is virtually certain. This likely increased risk as an instance of a common practice, although no entry made that afternoon could have had the firmware applied by 18:07.

Other findings

- It is very unlikely that the tide set down below prediction over Boolarong Bank. The gauge read 0.91 m at 18:12 against a predicted 0.94 m, and the vessel would have touched at the predicted height as well.
- A steering or propulsion failure before the vessel left the centreline is very unlikely. The recorded engine, hydraulic and rudder traces showed no fault, and every steering component tested on 12/07/2026 was serviceable.
- It is virtually certain that the engines were stopped and no attempt was made to refloat on a falling tide, and very likely that this reduced risk, because it kept a holed vessel with 38 passengers aboard on the bank rather than in 3.2 m of water.
- The unshielded boat ramp floodlights aligned behind the No. 4 beacon from the north are virtually certain to have existed. Whether they had any effect on the night cannot be determined either way.

Fictional case study for training. Not a real investigation.

- Both masters independently described the beacon as washed out against the new lights. For at least the last 37 seconds before the master called the beacon, it was clear of the islet and not seen.
- Against that, the beacon was screened by the canopy over most of the search, and the heading change began at the plotter before the search started. The master said twice that he could not tell whether the lights were why he could not see the beacon.
- The matter could not be resolved because no measurement exists of what could be seen from the vessel on the night, and no detection threshold is established for a 0.4 second green flash against that background. Both would be needed to resolve it.
- Whether the master was fatigued during the channel transit cannot be determined either way.
- In favour of fatigue, he was on day nine of a twelve-day tour, at the end of a split duty, after 6 hours 15 minutes of sleep, and he said he was tired.
- Against it, he said he was neither fighting to stay awake nor slow. The recorded performance agreed: the centreline was held within 3 m, the response to the shallow alarm took 4 seconds, and the radio work after the grounding was correct at the first attempt.
- No detailed fatigue assessment was made. Every decision that mattered on the night is either explained by another factor or contradicted by the recorded data, so even an established fatigue would not alter the factors above.

Safety action and recommendations

Clearing of the beacon's sightline deferred unfunded for three works rounds

Work order WO-118 was raised in 09/2023. It recorded mangrove regrowth on Merrandi Islet screening the No. 4 beacon from the northern approach, and it recommended trimming to 2 m on a three-yearly cycle. The work was deferred as unfunded at the 2023, 2024 and 2025 works rounds. The islet lies 240 m from the beacon, outside the 50 m scope of schedule 2.4. The work therefore competed for the general waterways vote rather than the aids to navigation vote. On the night of the grounding the canopy stood about 1.6 m above the 4.2 m lantern. Had the trim been funded in 2023, the canopy would very likely have stood about 0.7 m below the lantern, and the beacon would have been open to the channel between 0.84 nm and 0.34 nm, where it was screened.

The Harbour Master, Kanooka Heads, funded WO-176 from the 2026 waterways contingency, and the Merrandi Islet canopy was trimmed to 2 m on 24/07/2026. A three-yearly trimming cycle has been programmed, with the sightline re-measured from the inbound track on completion. This action restores the sightline at this site and answers the safety issue in part. It leaves unchanged the scope under which the work was deferred.

The 50 m vegetation scope at schedule 2.4 of the aids to navigation services agreement places work needed to keep an aid visible outside the aids to navigation vote whenever the obstruction lies further from the structure. It is recommended that [NOT ESTABLISHED: organisation to which the recommendation is addressed] replace that scope with a sightline test, so that work needed to keep an aid visible from the water it serves is funded as an aid to navigation whatever its distance from the structure.

Last service moved into darkness without reopening the daylight-only risk assessment

The route risk assessment for the Merrandi Channel, RRA-07 issue 2 (03/2023), restricted the route to daylight transits only. It required the General Manager Operations' approval and a specific passage plan for any night transit. Change record CHG-2026-014, approved by the Operations Manager on 22/04/2026, moved service TF18 from 16:15 to 17:35 for the winter timetable from 01/06/2026. The change was assessed as a timetable change only, with no new hazard identified, and a review of the route risk assessment was recorded as not required. RRA-07 was not reopened, no night passage plan was written, and the approval the restriction called for was neither sought nor given. TF18 then ran 31 times in darkness, and the occurrence was the thirty-first of those transits.

The significance of the issue lies in the control that was bypassed. The change process asked whether the change was new, not whether the existing assessment still held for the operation as changed. It is likely that a review of RRA-07 could not have closed without a night passage plan. Without that plan, the night transit had no hold point for loss of the primary fix. Why the change was assessed as a timetable change only was not established.

No safety action has been recorded in response to this issue, and no safety recommendation has been recorded for it.

No night passage plan or hold point for loss of the primary fix

The Merrandi Channel plan in appendix A to OPS-3 named the chart plotter as the primary means of position fixing. It named the visual transit of the No. 4 beacon and the town wharf leading lights

as the secondary means. It was written for the daylight transit and stated that night transits were not scheduled. There was no night passage plan for the channel. Neither the plan nor any other part of the manual said what a master should do if the plotter was lost during the transit, and the plan named no hold point or abort criterion. It is virtually certain that this gap existed. It very likely allowed the transit to continue at 8 knots from the plotter freeze until the shallow alarm.

This was the operator's own defence against this sequence, and it had never been written. A hold point would have cost nothing on a flat night with no other traffic. Both masters interviewed said independently that the right course was to stop, and both said that nothing told them to.

Safety action taken

[NOT ESTABLISHED: any safety action taken or begun by the operator in response to this safety issue]

Safety recommendations

Night passage plan with a hold point for the Merrandi Channel

The operator had no night passage plan for the Merrandi Channel and no hold point for loss of the primary means of position fixing.

It is recommended that the operator write a night passage plan for the Merrandi Channel before any further darkness transit. That plan should name a hold point. If the primary means of position fixing is lost inside the channel, the way is to be taken off the vessel until the position has been re-established by another means.

Defect procedure never requires a fault that cleared itself to be recorded

It is virtually certain that the operator's defect procedure, set out in OPS-9 of the Tarrakin Ferries Operations Manual, did not require a fault that had cleared itself to be recorded. Under 9.1, only a defect that rendered the vessel unseaworthy or affected a statutory item had to be reported immediately. The chart plotter was not a statutory item on this vessel on this route, although the passage plan named it as the primary means of position fixing. Under 9.2, any other defect went into the vessel defect book at the end of the shift in which it was found. For an intermittent fault that had already cleared, this provided no trigger at all. Under 9.3, the outgoing master briefed the relieving master on defects, but nothing in that brief had to be written down. Both masters independently understood the book to be for faults still present at handover. This explains why the plotter fault was passed on only by word of mouth at the change of master, but it does not excuse that omission.

The procedure mattered because it determined whether the plotter fault reached anyone ashore. None of the freezes in the preceding year was written down. Had they been recorded, they would have reached the Operations Manager's weekly review and the maintenance system, and from there the contractor. The contractor identified cause code 0x51 against the 11/2024 firmware in one bench test. It is likely that the absence of any record contributed to the freeze at 18:07. However, the operator held no subscription to the manufacturer's bulletins that carried the fix. It could not be established how many recorded freezes would have led to the unit being sent to the contractor, or how quickly.

No safety action has been recorded in response to this issue, and this report makes no recommendation on it.

Aids code set no notice or availability duty for beacons off declared channels

The Waterways Aids to Navigation Code WAN-2 places three duties on the responsible authority only where an aid stands on a declared shipping channel. The authority must meet an availability target and record availability. It must also give notice to mariners of a change to the aid's character, position or range. The Merrandi Channel is not a declared shipping channel. Its beacons therefore fall under section 4.3, which requires only that the responsible authority can demonstrate an inspection interval it has set itself. Neither that section nor the services agreement between the Kanooka Shire Council and the Kanooka Heads Port Authority requires availability to be recorded. Neither requires any user of the channel to be told of a defect, a deferral or a change.

It is unlikely that this gap influenced the grounding, for the reasons given in the analysis. It is a safety issue because of what it leaves in place. The channel carries a 96-passenger ferry in darkness. Yet the code gives the operator no right to be told when a beacon's range, character or position changes, and no measure of how often the aids work. The solar conversion of the No. 4 beacon in 11/2025 halved its nominal range, and for that reason it reached no one using the channel for eight months. It is likely that the gap warrants attention, because a future change to a beacon that is not screened would equally reach no one.

No safety action has been recorded in response to this safety issue, and this report makes no recommendation on it.

Beacon's range halved by solar conversion with no notice to channel users

The No. 4 beacon was converted from a mains supply to a solar lantern in November 2025 under the Council's waterways energy programme. The replacement lantern is rated at a nominal 3 nautical miles, against the 6 of the lantern it replaced, although the character, Fl G 2.5s, did not change. The investigator's measurement from the port workboat on 18 July 2026 gave a luminous range of 2.6 nm. A notification was lodged with the hydrographic service on 4 February 2026, and the chart correction was still pending on the day of the grounding. No local notice to mariners was issued, and no operator using the channel was informed. Tarrakin Ferries' passage plan appendix, issue 2, still recorded the beacon at 6 M.

The investigation found it unlikely that the change influenced this occurrence. It was nonetheless likely to have increased risk to the channel's users. A light on a 96-passenger route had its nominal range halved, and nobody using the channel was told. The operator's plan therefore described a light that was no longer there. On a night when the beacon is not screened, or after another aid is changed in the same way, a master working to the plan would be looking for a light twice as strong as the one shown.

On 22 July 2026 Kanooka Shire Council issued local notice to waterway users LNM-2026-14, giving the beacon's solar conversion and its 3 M nominal range. The Port Authority's aids to navigation register has been made available to every commercial operator using the channel. Tarrakin Ferries has reissued the passage plan appendix as issue 3 with the corrected range. This action is in progress and answers the safety issue in part. [NOT ESTABLISHED: which part of the safety issue remains unaddressed after this action.] No safety recommendation has been recorded for this issue.

Darkness transit rostered to close a split duty within 20 minutes of the spread limit

In the winter timetable, the TF18 darkness transit closes the afternoon half of a split duty. The master on that line works this duty on every day of a tour of 12 days on and 2 days off. The duty runs from 05:40 to 08:40 and from 13:00 to 19:20, which gives 9 hours 20 minutes of duty across a spread of 13 hours 40 minutes. That spread ends 20 minutes inside the operator's limit of 14 hours. No relief master is rostered, and a master who reports unfit is covered by extending another master's tour.

It was unlikely that this roster had any bearing on the grounding, because the recorded evidence showed no sign of fatigue in the master's performance on the night. It is nonetheless likely that the arrangement matters for future operations. It is a standing exposure built into the winter roster rather than a feature of a single day. Every winter darkness transit falls in the last hour and a quarter of a spread that runs close to the operator's own limit, on days nine to twelve of the tour as on the rest. The operator asks masters to keep a fatigue diary but does not read it, and there is no relief master to break a tour when a master reports unfit. This is a safety issue.

No safety action has been recorded in response to this issue, and no safety recommendation is made in respect of it.

Forward saloon lifejacket locker blocked by a 2019 counter fitted without re-survey

In the forward saloon of MV Tarrakin Star, a kiosk counter screened the lifejacket locker. The counter was fitted in a 2019 interior refit carried out under a minor modification notice, and that notice did not trigger a re-survey of the safety equipment arrangement. The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear.

The survey of 16/07/2026 found two problems with access:

- The locker could not be reached from 22 of the 34 forward saloon seats without passing behind the counter.
- With people standing at the counter, a 0.6 m inboard gap was the only route.

The muster placard directed passengers to a locker they could not see from their seats. The operator stated that the same counter is fitted to its two other 24 m vessels.

It was unlikely that this arrangement affected the grounding or its outcome. All 38 passengers were in lifejackets within about eight minutes, although the deckhand had to ask three passengers to move away from the counter to reach the locker. Even so, it is very likely that the arrangement increases the risk to passengers.

On the night, the vessel was aground, upright, stable and not sinking. Other circumstances could change that, such as a fire, a risk of capsizing, a night evacuation into the water, or a deckhand injured in an impact. In any of these, passengers who cannot reach their lifejackets without help from the crew would be left waiting for it. The locker remained behind the counter on this vessel at the time of the survey.

No safety action has been recorded in response to this issue, and no safety recommendation is made.

Distribution

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